

JAINENDRA KUMAR (JK) SINGH

SOLUTION LEADER | CLOUD MIGRATION & MODERNIZATION | DEVOPS & CONTAINERIZATION

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- Results-driven Cloud Solution Architect with 30+ years of IT experience and a decade of deep, hands-on Cloud practice. Proven track record designing and delivering large-scale cloud migration, application modernization, containerization, and DevOps solutions for Fortune 500 clients in healthcare, financial services, and government sectors.
- Adept at translating complex business requirements into scalable, secure AWS architectures — spanning multi-account governance, FinOps, disaster recovery, and container platforms.
- Brings measurable productivity gains to engineering teams and cloud operations by integrating GenAI-driven automation and AI-assisted development tooling into cloud architecture and delivery practices.

CORE COMPETENCIES

- **Cloud Architecture & Migration:** AWS Multi-Account Governance, Cloud Landing Zones, Application Modernization, Legacy Transformation, Hybrid & Multi-Cloud Strategy (AWS, Azure, GCP), Disaster Recovery & HA
- **DevOps & Platform Engineering:** CI/CD Pipelines, Containerization (Kubernetes, EKS, OpenShift), IaC (Terraform, CloudFormation), Docker, Helm, Service Mesh, Platform Engineering
- **Cloud Security & Compliance:** AWS Well-Architected Framework, Control Tower, SCPs, IAM, Security Hub, GuardDuty, FinOps & Cost Optimization, Cloud Governance
- **Enterprise Architecture:** Solution Architecture, Centers of Excellence, Architecture Governance, Reference Architecture Design, Data Migration & Analytics
- **GenAI & Automation:** GenAI/RAG Architectures, Agentic AI, MCP Servers, LLM Integration, AI-Assisted Development, AWS Bedrock
- **Leadership & Delivery:** Presales & RFP Leadership, Global Team Leadership, Agile Delivery, Budget & FinOps Management, Stakeholder Management, Business Development

PROFESSIONAL EXPERIENCE

HyLifen Chantilly, VA

Jul 2024 – Present

Solution Architect / Technology Consultant

Providing hands-on cloud architecture and GenAI consulting for Fortune 500 clients and growth-stage organizations in Healthcare, Financial Services, and Government sectors.

- Led container platform modernization for a large healthcare payer — designed migration solution for Pivotal Cloud Foundry workloads to Red Hat OpenShift, defining target topology, network policies, CI/CD pipelines, and operator patterns.
- Designed end-to-end AWS migration solution for 60 applications in a Healthcare M&A scenario — extended existing AWS environment using Terraform and CloudFormation automation, addressing compliance, security boundaries, and multi-account structure.
- Built Kubernetes cluster automation using AWS EC2 — scripted cluster bootstrapping and node management for a startup cloud environment, reducing setup time from days to hours.
- Architected a GenAI/RAG-based incident resolution platform for a major financial organization — ingesting mainframe, on-prem, and cloud knowledge bases to deliver AI-driven actionable insights to support teams.
- Established an AI-assisted development framework for a US Federal civilian agency — deployed Cursor and AWS GenAI tooling across 40 developers, achieving up to 5x developer productivity uplift.

Cognizant Technology Solutions College Station, TX
Director – Chief Cloud Architect (AWS Business Group)

Dec 2020 – Jun 2024

Spearheaded strategic AWS cloud initiatives, COE development, application modernization, and business growth for North American Fortune 500 accounts. Served as Chief Cloud Architect responsible for solution architecture, cloud Centers of Excellence, modernization frameworks, and CXO-level advisory across large-scale transformations, M&A programs, and regulated industry customers.

- Architected AWS-based solutions for 100+ critical applications during a major M&A program for a leading U.S. insurer — defined multi-account topology, network segmentation, IAM boundaries, security controls, and phased migration patterns.
- Defined enterprise AWS multi-account governance using AWS Organizations, Control Tower, and SCPs — standardized landing zones and infrastructure provisioning with Terraform and CloudFormation across 75+ accounts.
- Established Centers of Excellence in DB Migration, FinOps, Mainframe-to-Cloud, Containerization & Service Mesh — authored reference architectures and migration playbooks adopted enterprise-wide.
- Engineered cloud solutions to modernize legacy financial applications for a Fortune 100 company — enabling cloud-native adoption and measurable performance improvements.
- Designed CloudWatch alarm frameworks, dashboards, and performance metrics for key AWS services — improving observability and SLA adherence across customer environments.
- Co-created AWS security framework aligned with the AWS Well-Architected Framework — implementing industry security best practices across enterprise cloud environments.
- Owned solution architecture for 120+ enterprise RFPs spanning Cloud Migration, DevOps, FinOps, and Consulting — generating over \$25M in new cloud business within a year and driving a 50% increase in the AWS opportunity pipeline across 75 North American accounts.

Tata Consultancy Services (TCS) Multiple Locations, US

Nov 2002 – Dec 2020

Global Technology & Delivery Leadership | Cloud, Data & Enterprise Transformation

Held progressive senior leadership roles over an 18-year tenure with end-to-end responsibility for enterprise cloud architecture, platform modernization, and large-scale delivery programs. Served as executive delivery owner, client partner, and account head for strategic enterprise accounts — acting as the single point of accountability for architecture, delivery, commercial, and contractual matters.

- Served as account head and architecture lead for major enterprise clients including World Bank Group, Dow Chemical, Cardinal Health, Nielsen, Walgreens, Genworth, AXA, and CVS Caremark — spanning cloud migration, application modernization, data platform upgrades, and shared services transformation.
- Designed hybrid cloud strategy using AWS, VMware, and Nutanix for a Midwest utility — implemented Cloud Landing Zone as part of a \$2M Cloud-First milestone program. Delivered AWS cost optimization services producing up to 35% cost savings for banking and finance clients.
- Developed AWS-based microservices architecture for a major financial organization — leveraging DynamoDB, Lambda serverless routines, mainframe data replication, and Kafka streaming as part of a \$2.5M modernization program.
- Led cloud transformation and migration initiatives on AWS and Azure for World Bank Group digital platforms — managing end-to-end service delivery for SAP-based financial systems and leading executive steering committee meetings.
- Directed factory-model cloud migration of 2,000 applications to Azure for Dow Chemical — established migration patterns, governance, and wave planning at industrial scale across manufacturing environments.
- Led AWS S3-based data storage pilot and evaluated AWS solutions to replace IBM DataStage ETL platforms across 7,000 ETL jobs at Cardinal Health — overseeing engagements spanning ETL, ECM, Salesforce CRM, Dell Boomi, Hadoop, and SAP BO.
- Built and scaled global delivery organizations of 300–1,000+ professionals across India, Southeast Asia, LATAM, Europe, and the US — with full ownership of governance, cost, quality, and delivery outcomes.
- Defined and operationalized global operating models, governance structures, transition playbooks, and escalation mechanisms — enabling consistent multi-region delivery across all accounts.
- Partnered with global presales teams to shape transformation programs, managed services proposals, and RFP responses, driving long-term account growth initiatives.

Additional Experience (Syntel, Société Générale, HCL Perot)
Programmer Analyst | Project Manager | Various Roles

Aug 1995 – Nov 2002

Early career roles in software development, project management, and delivery across banking and technology sectors.

TECHNICAL PROFICIENCIES

- **AWS Core Services:** EC2, S3, RDS, VPC, IAM, Lambda, ECS, EKS, Fargate, Auto Scaling, ALB/NLB, CloudFront, Route 53, Direct Connect, CloudWatch
- **AWS Migration & Governance:** AWS Migration Hub, DMS, DataSync, Application Migration Service (MGN), AWS Organizations, Control Tower, SCPs, Config, Security Hub, GuardDuty, CloudTrail
- **AWS Data & Analytics:** DynamoDB, Aurora, Redshift, Glue, Lake Formation, Kinesis, MSK, Athena, EMR, Macie, Elasticsearch
- **AWS AI & ML:** Bedrock, Knowledge Bases, SageMaker, Textract, Comprehend, Rekognition, AWS Q
- **IaC & DevOps:** Terraform, CloudFormation, Jenkins, GitLab CI/CD, GitHub Actions, Boto3
- **Containers & Platform:** Kubernetes, Red Hat OpenShift, EKS, Docker, Helm, Service Mesh (Istio), CI/CD Pipelines
- **Multi-Cloud:** Azure (AI Foundry, AKS, Data Factory, Synapse, Databricks) | GCP (Vertex AI, GKE, BigQuery)
- **GenAI & AI Platforms:** AWS Bedrock | LangChain | LangGraph | RAG Architectures | Agentic AI | MCP Servers | OpenAI | Ollama | Hugging Face
- **Programming Languages:** Python | JavaScript / Node.js | SQL | Java | C/C++
- **Databases:** Oracle | SQL Server | DynamoDB | PostgreSQL | MySQL | MongoDB | DB2
- **Observability:** CloudWatch | OpenTelemetry | Datadog | Grafana

EDUCATION, PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

- Bachelor of Engineering (BE), Electrical & Electronics — National Institute of Technology Karnataka, India
- Chief Technology Officer (CTO) Program — The Wharton School, University of Pennsylvania
- Scaling a Business: How to Build a Unicorn — The Wharton School, University of Pennsylvania
- Executive Presence and Influence — The Wharton School, University of Pennsylvania

Certifications:

- AWS Solutions Architect – Associate
- AI Security & Governance Certified
- Generative AI Fundamentals — Databricks
- Oracle OCI 2025 AI Foundations Associate
- Career Essentials in Generative AI — Microsoft & LinkedIn
- Model Context Protocol Advanced Topics — Anthropic | AI Agents Fundamentals — Hugging Face
- Healthcare Delivery Providers & Healthcare Marketplace — University of Minnesota (Coursera)